2 (a) (i) Electric field between parallel plates, $E = \dfrac{V}{d}$

$$= \frac{600-(-600)}{0.025}$$

$$= 4.8 \times 10^{4}\ \text{V m}^{-1}$$

(ii) Work done W = change in potential energy U

$$W = \Delta U = q\Delta V$$

$$= (8.0 \times 10^{-19})(500 - (-400))$$

$$= 7.2 \times 10^{-16}\ \text{J}$$

(b) The electric field strength between parallel plates is uniform.

Since electric field strength is the negative of the potential gradient ($E = -\dfrac{dV}{dr}$), the potential decreases **uniformly** from the left plate (+600 V) to the right plate (-600 V).

Therefore, the central equipotential line, which is midway between the +600 V and -600 V plates, must be the zero-volt equipotential line.

Note: Alternatively, we can explain using definition of electric potential. In the absence of other charged bodies, the two work done in bringing a charge from infinity to the central position due to the electric potential of each plate are equal in magnitude but opposite in sign. Hence the potential along the central line is zero.